MonkeyOCR: Document Parsing with a *Structure-Recognition-Relation* Triplet Paradigm

Zhang Li[1], Yuliang Liu[1,†], Qiang Liu[2], Zhiyin Ma[1], Ziyang Zhang[1], Shuo Zhang[1], Zidun Guo[1], Jiarui Zhang[2], Xinyu Wang[1], Xiang Bai[1]

Final Project Presentation | *December 2025*

Group 3:

- *F11315101* Aldo Acevedo
- *F11315102* Alejandro Adorno
- *F11315107* Fabrizio Diaz

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Deep Learning For Computer Vision
Prof. G.S. Jison Hsu 徐繼聖 教授
Artificial Vision Laboratory

Part I: *Problem Statement*

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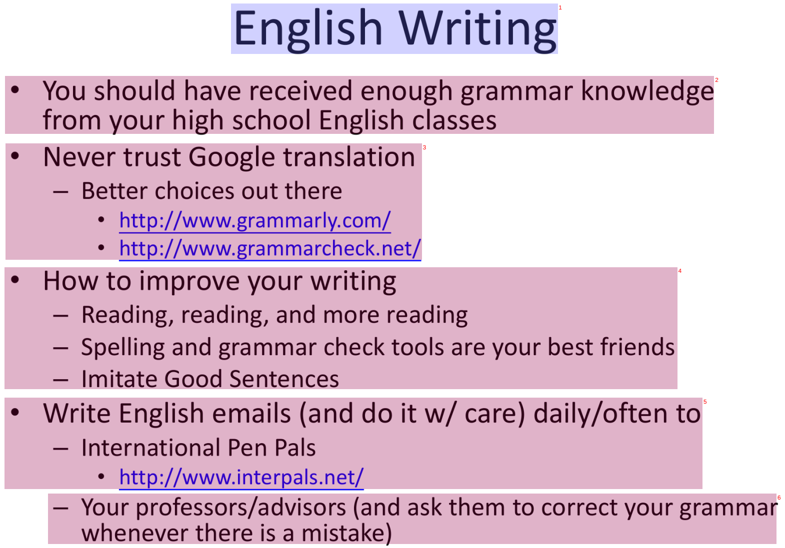
Artificial Vision Laboratory

The Document Parsing Challenge

- **OCR(*Optical Character Recognition*)** is a technology that converts images of text (scanned documents, photos, PDFs) into editable, machine-readable text.
- ***Document parsing*** is OCR + understanding layout, structure, tables, formulas, and how everything relates in order to generate structured, machine-readbale information.



$\Rightarrow$

```
# English Writing

• You should have received enough grammar knowledge from your
high school English classes

• Never trust Google translation
- Better choices out there
• http://www.grammarly.com/
• http://www.grammarcheck.net/

• How to improve your writing
– Reading, reading, and more reading
– Spelling and grammar check tools are your best friends
– Imitate Good Sentences

• Write English emails (and do it w/ care) daily/often to
– International Pen Pals
• http://www.interpals.net/

– Your professors/advisors (and ask them to correct your
grammar whenever there is a mistake)
```

Part II: *OmniDocBench Benchmark*

OmniDocBench Overview

What is OmniDocBench?

- **Standard benchmark** for evaluating document parsing models
- **981 PDF pages** with **100k+ human annotations**
- Covers **9 document types**: academic papers, textbooks, slides, newspapers, financial reports, exam papers, books, magazines, handwritten notes
- **4 layout styles**: single-column, double-column, three-column, complex
- **3 language categories**: English, Chinese, Mixed

Key Metrics

- **Edit Distance** ↓ (text): character-level errors — insertions, deletions, substitutions
- **TEDS** ↑ (tables): Tree-Edit-Distance Similarity for table structure + content
- **CDM** ↑ (formulas): Character Detection Matching for LaTeX
- **Overall** ↑ : weighted Edit Distance across all content types

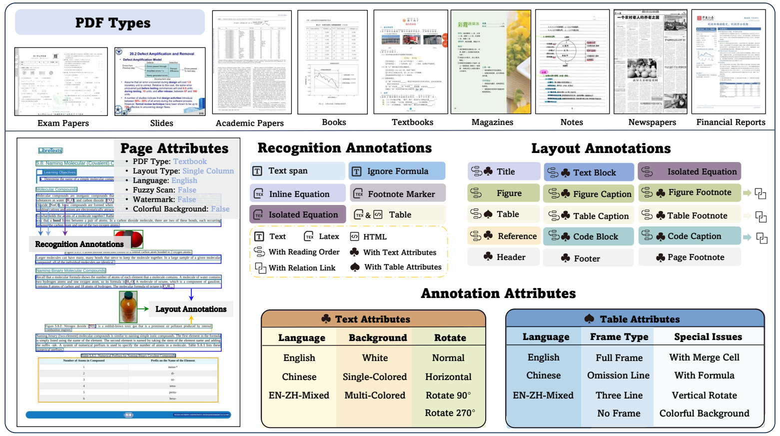


Figure 1: *Overview of OmniDocBench Data Diveristy*

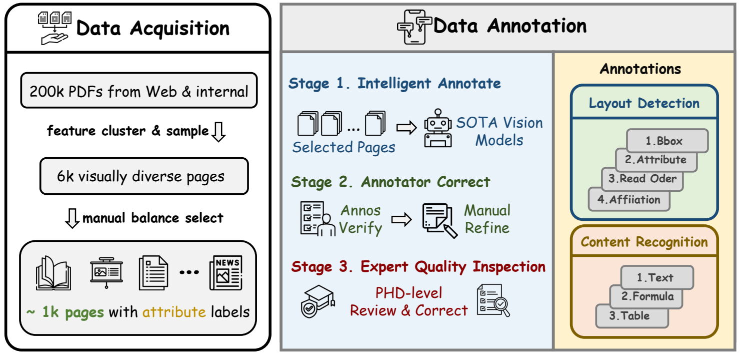
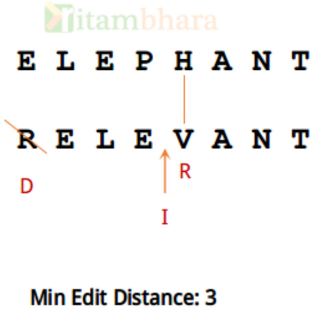


Figure 2: *Overview of the OmniDocBench dataset construction*

Text Block Edit Distance

		E	L	E	P	H	A	N	T
	0	1	2	3	4	5	6	7	8
R	1	1	2	3	4	5	6	7	8
E	2	1	2	2	3	4	5	6	7
L	3	2	1	2	3	4	5	6	7
E	4	3	2	1	2	3	4	5	6
V	5	4	3	2	2	3	4	5	6
A	6	5	4	3	3	3	3	4	5
N	7	6	5	4	4	4	4	3	4
T	8	7	6	5	5	5	5	4	3



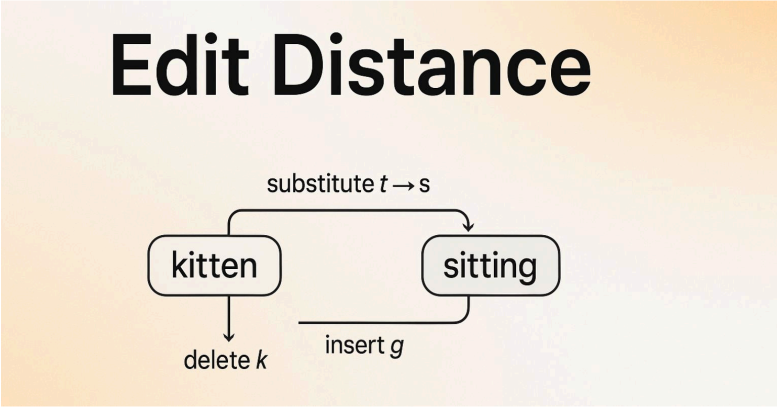


Figure 3: *Overview of the Normalized Edit Distance metric*

$$\text{Edit Distance Score} = \left(1 - \frac{\text{Levenshtein Distance}}{\max(\text{len(pred)}, \text{len(ref)})}\right)$$

- **Normalized Metric:** Quantifies character-level accuracy using Levenshtein distance (insertions, deletions, substitutions), normalized by the maximum string length.
- **Dynamic Alignment:** Utilizes an **Adjacency Search Match** algorithm to intelligently merge or split paragraphs, ensuring optimal alignment between prediction and ground truth before scoring.
- **Unified Scope:** Evaluates "pure text" by integrating code blocks and converting inline formulas to standard Unicode, treating them as part of the textual flow.
- **Noise Filtration:** Applies rigorous preprocessing to strip markdown tags and remove images, isolating the semantic text content for a fair comparison.

Table TEDS

Predicted

```
 <table>
<tr><td>Name</td><td>Score</td></tr>
<tr><td>Alice</td>95</td></tr>
 </table>
```

Ground Truth

```
 <table>
<tr><td>Name</td><td>Score</td></tr>
<tr><td>Alice</td><td>95</td></tr>
 </table>
```

Listing 1: **TEDS SCORE:** 85% (partial structural match, missing one <td> element

$$\text{Table TEDS} = \left(1 - \frac{\text{Tree Edit Distance}}{\max\left(|\text{tree}_{\text{pred}}|, |\text{tree}_{\text{ref}}|\right)}\right)$$

- **Structure-Aware Metric:** Unlike simple text overlap, TEDS evaluates both the **logical structure** (HTML tags) and **cell content** simultaneously.
- **Tree Representation:** The table is converted into a DOM tree (e.g., `<table>` → `<tr>` → `<td>`), enabling detection of complex errors like spanning cell alignment.
- **Edit Operations:** It calculates the minimum number of **insertions**, **deletions**, and **substitutions** of nodes needed to transform the Predicted tree into the Ground Truth tree.
- **Holistic Score: - Structural Edits:** Penalizes missing rows, columns, or headers.

Formula CDM

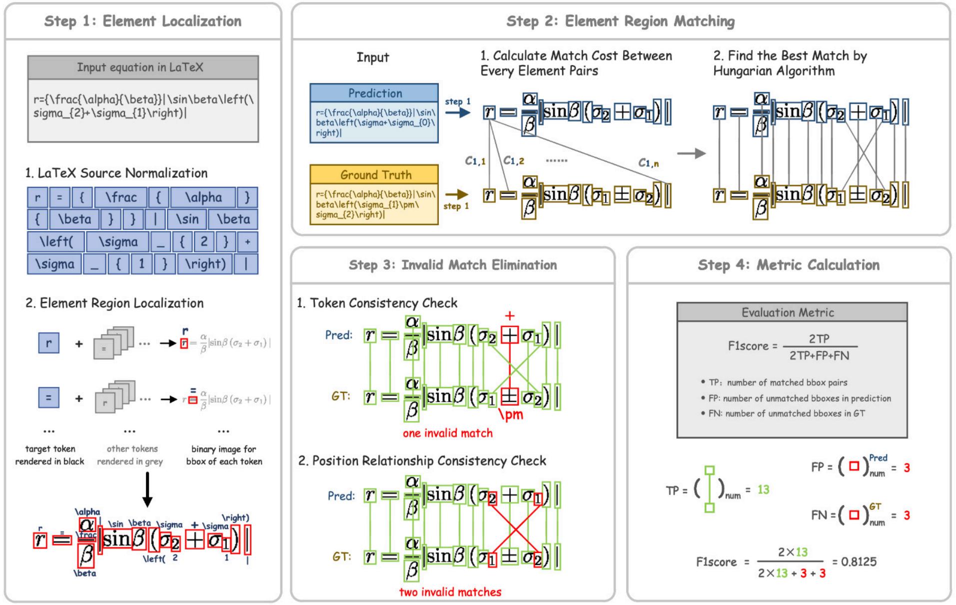


Figure 4: *Overview of the Character Detection Matching (CDM)*

$$\mathrm{CDM} = \frac{2 \times \mathrm{TP}}{2 \times \mathrm{TP} + \mathrm{FP} + \mathrm{FN}}$$

- **Step 1 (Localization):** We first "normalize" the LaTeX and render it to a binary image to find exactly where each character (like α or β) sits on the page.
- **Step 2 (Matching):** We calculate the "cost" of matching a character in the Prediction to a character in the Ground Truth.
- **Step 3 (Elimination):** We discard "bad" matches.
- **Step 4 (Calculation):** The final CDM score is essentially an F1-score.

Overall

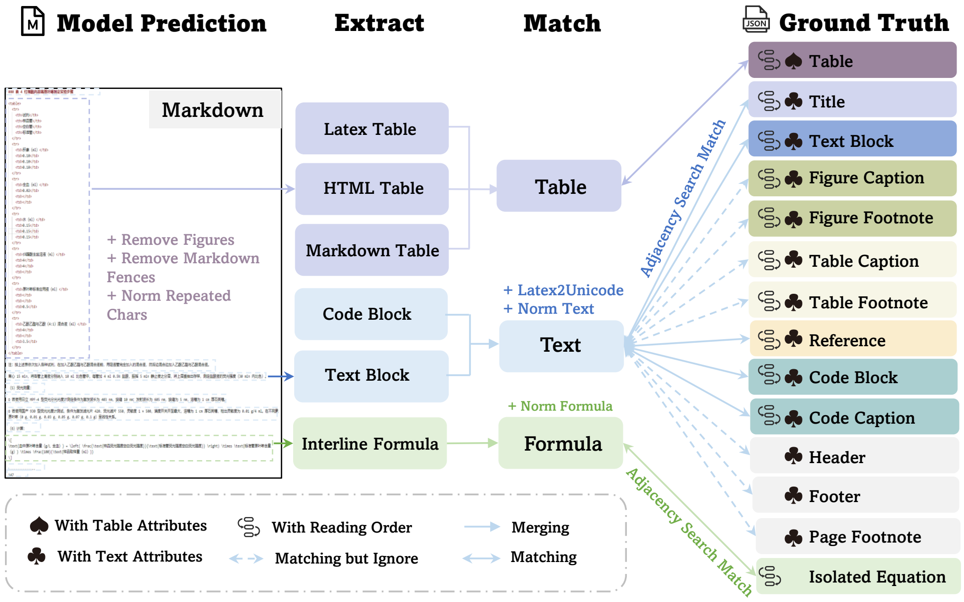


Figure 5: *OmniDocBench Evaluation Pipeline*

$$\text{Overall} = \frac{(1 - \text{Text Edit Distance} \times 100 + \text{Table TEDS} + \text{Formula CDM})}{3}$$

- **Composite Metric:** Averages scores from Text, Tables, and Formulas for a holistic performance indicator.
- **Text Conversion:** Inverts Edit Distance $((1 - \text{Edit\_dist}) \times 100)$ to quantify text accuracy rather than error rate.
- **Structure-Aware:** Incorporates TEDS to evaluate both the hierarchical structure and cell content of tables.
- **Visual Robustness:** Utilizes CDM for formulas to assess visual correctness, handling complex LaTeX variations.
- **Balanced Assessment:** Weights complex elements equally with plain text to identify bottlenecks across all content types.

Part III: *MonkeyOCR*

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Paper Approach at *Document Parsing*

End-to-End Approaches

Slow because they process entire pages at once, making them impractical for large-scale applications

Fatal Flaw: *Computational Bottleneck*

- ❌ $O(n^2)$ attention complexity
- ❌ Long input/output sequences
- ❌ 5-7× slower than pipelines

Structure-Recognition-Relation (SRR) Triplet Paradigm

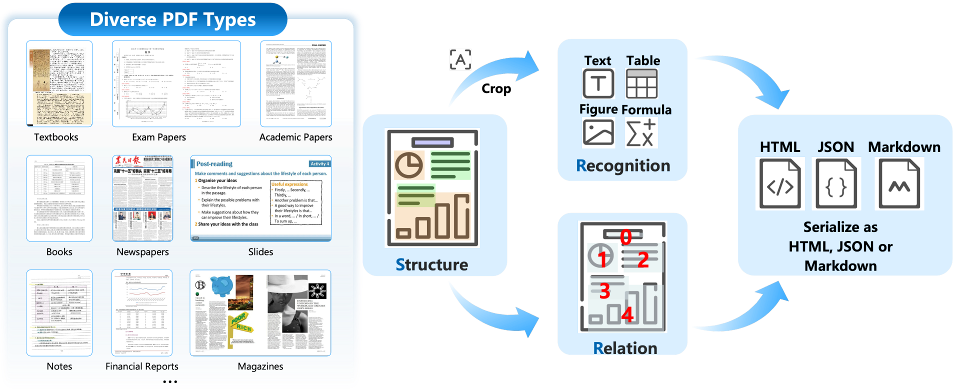


Figure 6: **MonkeyOCR** approach at *Document Parsing*

Pipeline-based Approaches

Error propagation across sequential stages - mistakes in early detection compound in later recognition

Fatal Flaw: *Error Propagation*

- ❌ Early mistakes compound downstream
- ❌ Formula detection crops text → OCR fails

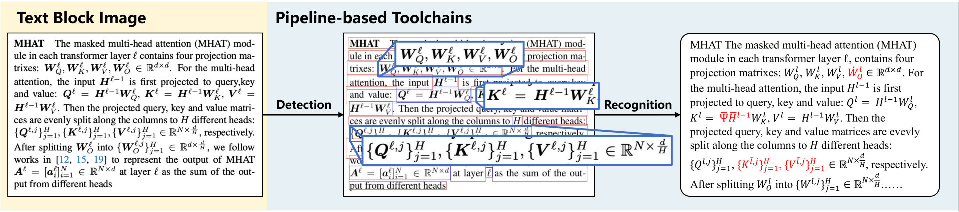


Figure 7: *Example of error propagation in pipeline-based methods*

1. "*Where is it?*" → Structure detection (YOLO-based layout analysis)
2. "*What is it?*" → Recognition (unified LMM for parallel block-level content extraction)
3. "*How is it organized?*" → Relation prediction (reading order model)

Models Used

Structure

PP-DocLayout

Relation

LayoutReader

Recognition

Qwen2.5VL - No specifications or details

Questions Left Unanswered:

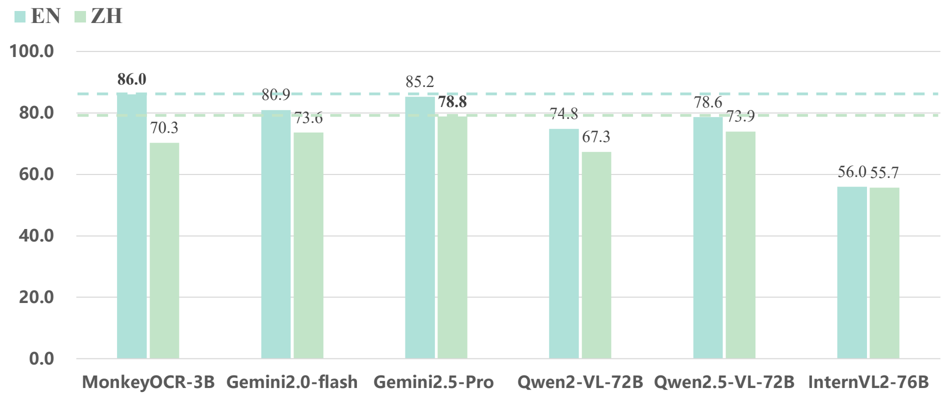
- Which checkpoint version? (e.g., `Qwen2.5-VL-7B-Instruct`, `Qwen2.5-VL-4B-Instruct`)
- What quantization or compression was applied?
- Were any custom modifications made to the base model besides fine-tuning?
- How does the choice of backbone affect downstream performance?

Our Approach: This ambiguity motivated us to explore alternative models and systematically evaluate different configurations.

MonkeyOCR

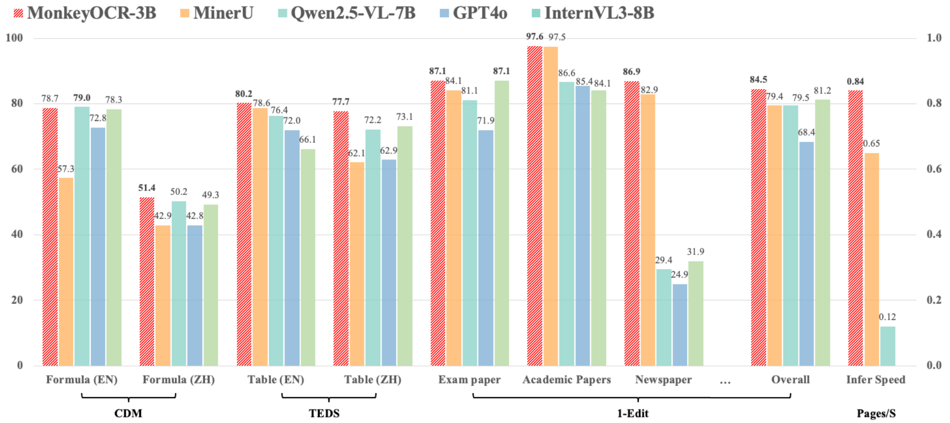
MonkeyOCR-3B Details

- *Backbone:* Qwen2.5-VL
- *Optimizer:* AdamW with learning rate $2e^{-5}$, cosine schedule
- *Training:* Batch size 64, 53 hours on 32×A800 GPUs
- *Deployment:* Runs on single RTX 3090 GPU via LMDeploy



Training Data

- ***MonkeyDoc:*** 3.9M instances across all parsing tasks
- 10+ document types (papers, textbooks, newspapers, etc.)
- Bilingual support (Chinese + English)



Results: Baseline

	text_block_Edit_dist_score	display_formula_CDM	table_TEDS	table_TEDS_structure_only	overall
MonkeyOCR-3B (baseline)	92.559	84.070	70.763	74.704	82.464000

Part IV: *Experiments*

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What can we improve on?

Option 1: Benchmark Performance

Focus on improving scores on OmniDocBench across all metrics

- Text Edit Distance
- Table TEDS
- Formula CDM
- Overall composite score

Option 2: Model Compression

Reduce model size through:

- Quantization
- Knowledge distillation
- Pruning

Option 3: Speed Optimization

Increase inference speed by:

- Selecting lighter backbone
- Modifying architecture
- Optimizing code paths
- Reducing computation

Recognition backbone upgrade

Initial Direction: Exchange recognition backbone

Why Qwen3-VL?

- ✓ Newer architecture (better base performance)
- ✓ Improved efficiency (fewer parameters)
- ✓ Better multilingual support
- ✓ Enhanced visual understanding
- ✓ Compatible API (drop-in replacement)

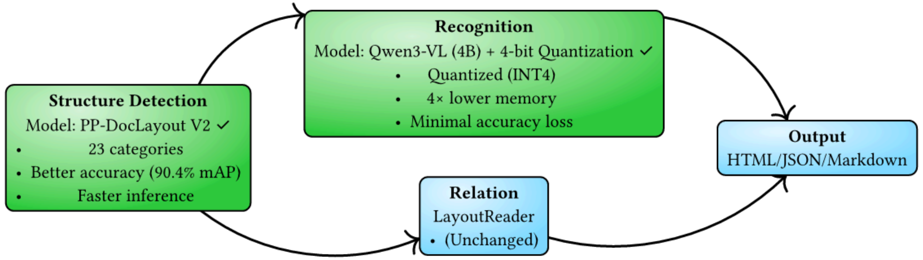


Figure 8: MonkeyOCR with upgraded backbone

Implementation

- Replace `Qwen2.5-VL` with `Qwen3-VL-Instruct-2B`
- Keep MonkeyOCR architecture unchanged
- No code refactoring required

Advantage: Lowest-risk, highest-reward change, testing a single variable

Model Comparison

- **Qwen2.5-VL:** Base architecture
- **Qwen3-VL:** Enhanced version
  - Faster inference
  - Better accuracy
  - Smaller footprint

Results: Recognition backbone upgrades

	text_block_Edit_dist_score	display_formula_CDM	table_TEDS	table_TEDS_structure_only	overall
MonkeyOCR-3B (baseline)	92.559000	84.070000	70.763000	74.704000	82.464000
InternVL3.5-1B-Flash	75.097000	38.145000	71.851000	81.624000	61.697667
Qwen3-VL-2B-Instruct	90.942000	80.677000	73.437000	79.425000	81.685333
Qwen3-VL-2B-Instruct-bnb-4bit	91.172000	82.096000	71.506000	78.633000	81.591333

Error Analysis

Key Insight: While text recognition improved, the model's lack of comfort with structured content (tables, formulas) reveals a fundamental limitation, specially in Chinese. This motivated deeper investigation into why structured parsing failed. After upgrading to Qwen3-VL-Instruct-2B, we evaluated performance on OmniDocBench and discovered unexpected weaknesses:

Table TEDS Degradation

Model struggles with:

- Table structure output
- Cell hierarchy understanding in Chinese
- Relationship between elements

Issue: Model not "comfortable" with structural HTML layouts

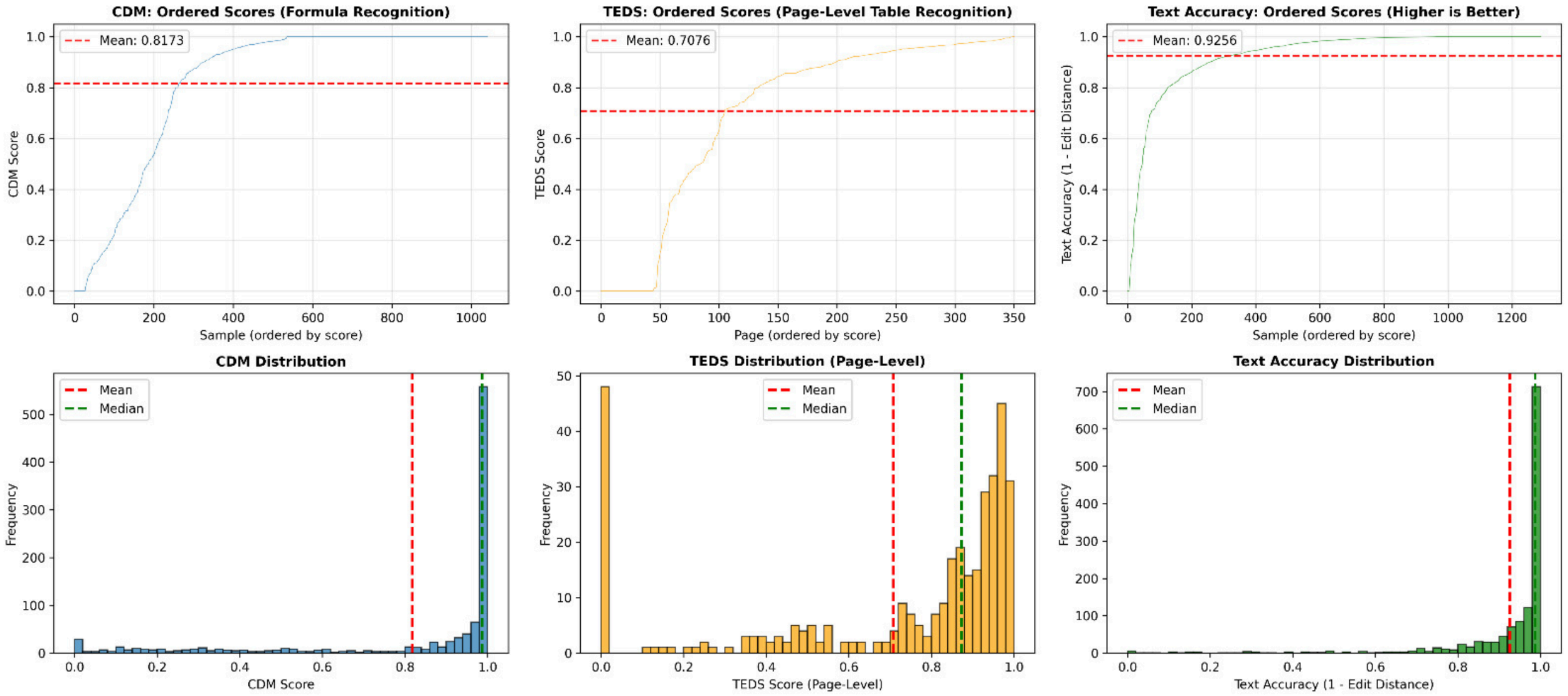
Formula CDM Degradation

Model struggles with:

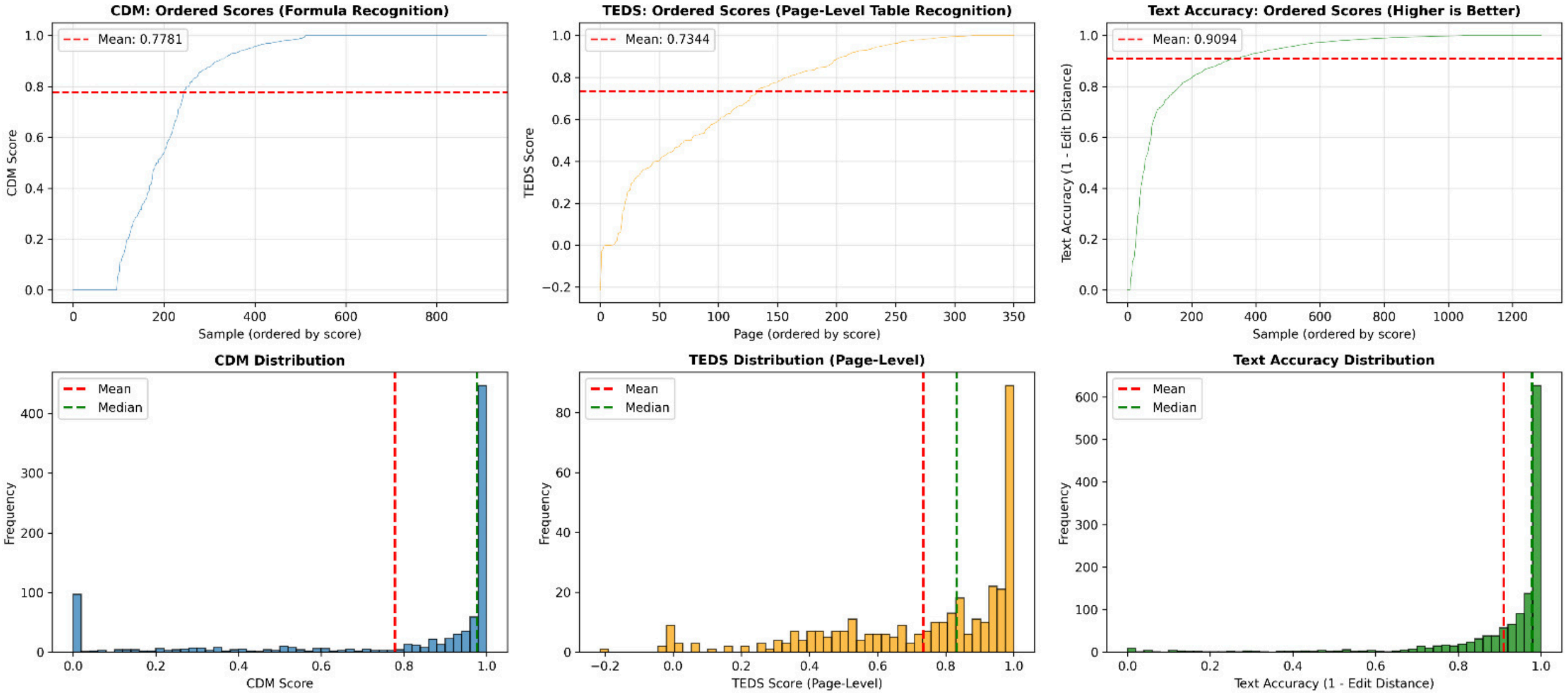
- LaTeX formula output
- Mathematical notation parsing in Chinese
- Visual formula representation

Issue: Model lacks structural understanding of LaTeX formulas

Error Analysis – Error distribution (MonkeyOCR-3B)

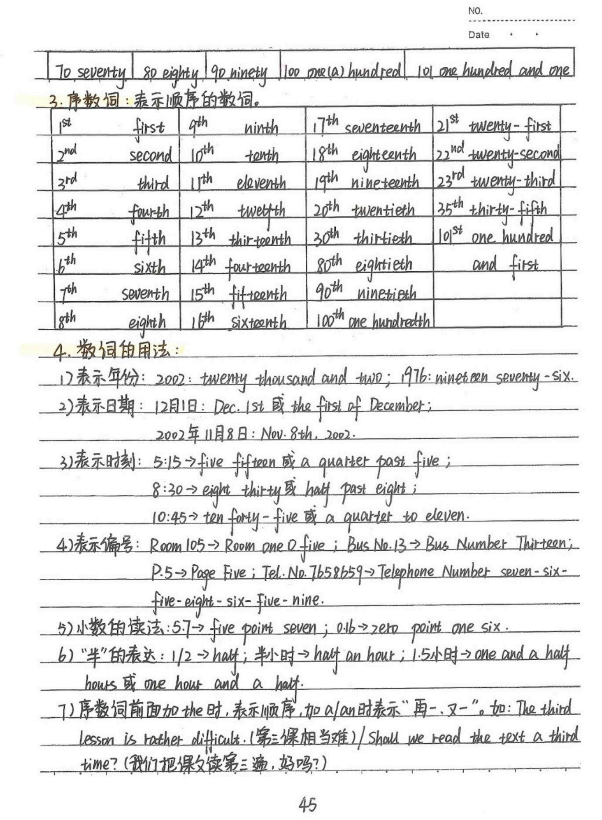


Error Analysis – Error distribution (Qwen3-VL-2B)



Error Analysis – Example 1 – Table

Image


NO.
Date

70 seventy	80 eighty	90 ninety	100 one(a) hundred	101 one hundred and one

3. 序数词：表示顺序的数词。

1^{st} first	9^{th} ninth	17^{th} seventeenth	21^{st} twenty-first
2^{nd} second	10^{th} tenth	18^{th} eighteenth	22^{nd} twenty-second
3^{rd} third	11^{th} eleventh	19^{th} nineteenth	23^{rd} twenty-third
4^{th} fourth	12^{th} twelfth	20^{th} twentieth	35^{th} thirty-fifth
5^{th} fifth	13^{th} thirteenth	30^{th} thirtieth	101^{st} one hundred
6^{th} sixth	14^{th} fourteenth	80^{th} eightieth	and first
7^{th} seventh	15^{th} fifteenth	90^{th} ninetieth	
8^{th} eighth	16^{th} sixteenth	100^{th} one hundredth	

4. 数词的用法：
1) 表示年份：2002：twenty thousand and two；1976：nineteen seventy-six.
2) 表示日期：12月1日：Dec. 1st 或 the first of December；
2002年11月8日：Nov. 8th, 2002.
3) 表示时刻：5:15 → five fifteen 或 a quarter past five；
8:30 → eight thirty 或 half past eight；
10:45 → ten forty-five 或 a quarter to eleven.
4) 表示编号：Room 105 → Room one O five；Bus No. 13 → Bus Number Thirteen；
P.5 → Page Five；Tel. No. 7658659 → Telephone Number seven-six-five-eight-six-five-nine.
5) 小数的读法：5.7 → five point seven；0.16 → zero point one six.
6) "半"的表达：1/2 → half；半小时 → half an hour；1.5小时 → one and a half hours 或 one hour and a half.
7) 序数词前面加the时，表示顺序，加a/an时表示"再一、又一"。如：The third lesson is rather difficult.(第三课相当难)/Shall we read the text a third time?(我们把课文读第三遍，好吗？)

45

Ground truth

```
<html><body><table border="1" ><tr><td>$1^{st}$
first</td><td>$9^{th}$ ninth</td><td>$17^{th}$
seventeenth</td><td>$21^{st}$ twenty-first</td></
tr><tr><td>$2^{nd}$ second</td><td>$10^{th}$
tenth</
td><td>$18^{th}$ eighteenth</td><td>$22^{nd}$
twenty-second</td></tr><tr><td>$3^{rd}$ third</
td><td>$11^{th}$ eleventh</td><td>$19^{th}$
nineteenth</td><td>$23^{rd}$ twenty-third</td></
tr><tr><td>$4^{th}$ fourth</td><td>$12^{th}$
twelfth</
td><td>$20^{th}$ twentieth</td><td>$35^{th}$
thirty-fifth</td></tr><tr><td>$5^{th}$ fifth</
td><td>$13^{th}$ thirteenth</td><td>$30^{th}$
thirtieth</td><td>$101^{st}$ one hundred </
td></tr><tr><td>$6^{th}$ sixth</td><td>$14^{th}$
fourteenth</td><td>$80^{th}$ eightieth</
td><td>and first</td></
tr><tr><td>$7^{th}$ seventh</td><td>$15^{th}$
fifteenth</td><td>$90^{th}$ ninetieth</td><td></
td></tr><tr><td>$8^{th}$ eighth</td><td>$16^{th}$
sixteenth</td><td>$100^{th}$ one hundredth</
td><td></td></tr></table></body></html>
```

Predicted

```
<html><body><table border="1" ><tr><td></
td><td>first</td><td>9th</td><td>ninth</
td><td>17th</td><td>seventeenth</td><td>21st</
td><td>twenty-first</td></tr><tr><td>2nd</
td><td>second</td><td>10th</td><td>tenth</
td><td>18th</td><td>eighteenth</td><td>22nd</
td><td>twenty-second</td></tr><tr><td>3rd</
td><td>third</td><td>11th</td><td>eleventh</
td><td>19th</td><td>nineteenth</td><td>23rd</
td><td>twenty-third</td></tr><tr><td>4th</
td><td>fourth</td><td>12th</td><td>twelfth</
td><td>20th</td><td>twentieth</td><td>35th</
td><td>thirty-fifth</td></tr><tr><td>5th</
td><td>fifth</td><td>13th</td><td>thirteenth</
td><td>30th</td><td>thirtieth</td><td>101st</
td><td>one hundred</td></tr><tr><td>6th</
td><td>sixth</td><td>14th</td><td>fourteenth</
td><td>80th</td><td>eightieth</
td><td></td><td>and first</td></tr><tr><td>7th</
td><td>seventh</td><td>15th</td><td>fifteenth</
td><td>90th</td><td>ninetieth</td><td></td><td></
td></tr><tr><td>8th</td><td>eighth</td><td>16th</
td><td>sixteenth</td><td>100th</td><td>one
hundredth</t
```

cut-off for readability........

Error Analysis – Example 2 – Formula

Image

$$
\begin{aligned}
\sum_{n=1}^{\infty} \frac{2^{\langle n\rangle}+2^{-\langle n\rangle}}{2^n} &= \sum_{k=1}^{\infty} \sum_{n,\langle n\rangle=k} \frac{2^{\langle n\rangle}+2^{-\langle n\rangle}}{2^n} \\
&= \sum_{k=1}^{\infty} \sum_{n=k^2-k+1}^{k^2+k} \frac{2^k+2^{-k}}{2^n} \\
&= \sum_{k=1}^{\infty} (2^k+2^{-k})(2^{-k^2+k}-2^{-k^2-k}) \\
&= \sum_{k=1}^{\infty} (2^{-k(k-2)}-2^{-k(k+2)}) \\
&= \sum_{k=1}^{\infty} 2^{-k(k-2)} - \sum_{k=3}^{\infty} 2^{-k(k-2)} \\
&= 3.
\end{aligned}
$$

Ground truth

```
{rl}
\sum_{n=1}
^{\infty}\frac{2^{\langlen\rangle}+2^{-
\langlen\rangle}}{2^n}&=\sum_{k=1}
^{\infty}\sum_{n,
\langlen\rangle=k}\frac{2^{\langlen\rangl
\langlen\rangle}}{2^n}\\\\
&=\sum_{k=1}^{\infty}\sum_{n=k^2-k+1}
^{k^2+k}\frac{2^k+2^{-k}}{2^n}\\\\
&=\sum_{k=1}^{\infty}(2^k+2^{-k})
(2^{-k^2+k}-2^{-k^2-k})\\\\
&=\sum_{k=1}^{\infty}(2^{-
k(k-2)}-2^{-k(k+2)})\\\\
&=\sum_{k=1}^{\infty}2^{-k(k-2)}-
\sum_{k=3}^{\infty}2^{-k(k-2)}\\\\
&=3.
```

Predicted

```
\sum_{n=1}^{\infty}\frac{2^{<n>}+2^{-
<n>}}{2^n}=\sum_{k=1}
^{\infty}\sum_{n,<n>=k}\frac{2^{(n)}+2^{-
<n>}}{2^n}=\sum_{k=1}
^{\infty}\frac{k^2+k}{k^2-
k+1}\frac{2^k+2^{-k}}{2^n}=\sum_{k=1}
^{\infty}(2^k+2^{-k})(2^{-k^2+k}-2^{-
k^2-k})=\sum_{k=1}^{\infty}(2^{-
k(k-2)}-2^{-k(k+2)})=3a=1
```

LoRA

LoRA is a parameter-efficient fine-tuning technique that avoids updating all model weights.

Core Idea

Instead of modifying pretrained weights W_0, we add a low-rank decomposition:

$$W = W_0 + \Delta W = W_0 + AB^T$$

where:

- $W_0 \in R^{\{d\times k\}}$ = pretrained weights (frozen)
- $A \in R^{\{d\times r\}}$ = down-projection (learnable)
- $B \in R^{\{k\times r\}}$ = up-projection (learnable)
- r = rank (typically small: 8-64)

Advantages

- ✓ **Massive parameter reduction:** Only train (r(d+k)) params instead of (d times k)
- ✓ **Memory efficient:** 10-100× less VRAM needed
- ✓ **Faster training:** Fewer gradients to compute
- ✓ **Modularity:** Can swap adapters between tasks

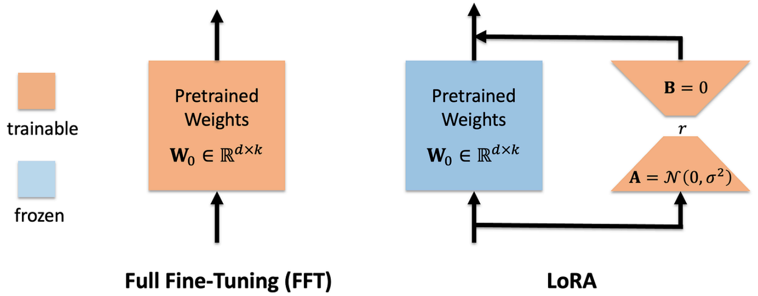


Figure 9: Full Fine-Tuning vs. LoRA Architecture

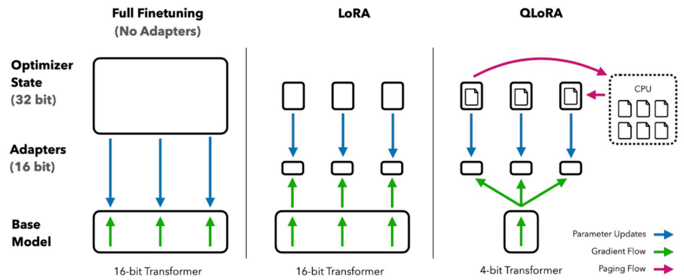


Figure 10: Efficiency Comparison: FFT, LoRA, QLoRA

PubTabNet Dataset

Dataset Statistics

- **Total samples:** 568,627 table images
- **Training set:** 515,679 tables
- **Validation set:** 52,948 tables
- **Our usage:** 15,000 samples for fine-tuning
- **Source:** arXiv papers (PDFs)

Table Characteristics

- Diverse layouts (simple to complex)
- Multiple column/row spans
- Varying table sizes
- Different fonts and styles
- Real-world document quality

Task Focus

- HTML/markdown structure generation
- Cell content extraction
- Row/column relationship preservation
- Hierarchical table representation

PubTabNet Dataset

Rank	Lane	Name	Nationality	Time	Notes
	4	Michael Phelps	United States	51.25	OR
	3	Ian Crocker	United States	51.29	
	5	Andriy Serdinov	Ukraine	51.36	EU
4	1	Thomas Rupprath	Germany	52.27	
5	6	Igor Marchenko	Russia	52.32	
6	2	Gabriel Mangabeira	Brazil	52.34	
7	8	Duje Draganja	Croatia	52.46	
8	7	Geoff Huegill	Australia	52.56	

```
<table border="1" cellspacing="0">
<tr><th>Rank </th> <th> Lane </th> <th> Name </th> <th> Nationality
</th> <th> Time </th> <th> Notes </th> </tr> <tr> <td> </td> <td> 4
</td> <td> Michael Phelps </td> <td> United States </td> <td> 51.25
</td> <td> OR </td> </tr> <tr> <td> </td> <td> 3 </td> <td> Ian
Crocker </td> <td> United States </td> <td> 51.29 </td> <td> </td> </
tr> <tr> <td> </td> <td> 5 </td> <td> Andriy Serdinov </td> <td>
Ukraine </td> <td> 51.36 </td> <td> EU </td> </tr> <tr> <td> 4 </td>
<td> 1 </td> <td> Thomas Rupprath </td> <td> Germany </td> <td> 52.27
</td> <td> </td> </tr> <tr> <td> 5 </td> <td> 6 </td> <td> Igor
Marchenko </td> <td> Russia </td> <td> 52.32 </td> <td> </td> </tr>
<tr> <td> 6 </td> <td> 2 </td> <td> Gabriel Mangabeira </td> <td>
Brazil </td> <td> 52.34 </td> <td> </td> </tr> <tr> <td> 7 </td> <td>
8 </td> <td> Duje Draganja </td> <td> Croatia </td> <td> 52.46 </td>
<td> </td> </tr> <tr> <td> 8 </td> <td> 7 </td> <td> Geoff Huegill </
td> <td> Australia </td> <td> 52.56 </td> <td> </td> </tr> </table>
```

Figure 11: PubTabNet Example: Table image with HTML structure

SynthFormulaNet Dataset

Dataset Statistics

- **Total samples:** 100,000+ synthetic formulas
- **Formula complexity:** Basic to advanced calculus
- **Rendering variation:** Multiple fonts and sizes
- **Augmentations:** Rotation, noise, degradation
- **LaTeX format:** Ground truth annotations

Why SynthFormulaNet: Synthetic data provides controlled, diverse formula variations essential for robust mathematical content parsing.

Formula Types Covered

- Inline equations
- Display equations
- Multi-line expressions
- Greek letters and symbols
- Superscripts and subscripts
- Fractions and matrices

Task Focus

- LaTeX source generation
- Character-level recognition
- Mathematical notation understanding
- Handling complex expressions

SynthFormulaNet Dataset

$$\frac{d}{ds}G(s) = -TrA(s)^{s}\sum_{i=1}^{a}\pi_i S_i^{\frac{1}{1+s}}\left[\log S_i^{\frac{1}{1+s}} - \log A(s)\right] . \qquad (25)$$

Figure 12: SynthFormulaNet Example: Formula image with LaTeX

```
<loc_0><loc_0><loc_500><loc_500>\\frac { d } { d s } G ( s ) = - T r
A ( s ) ^ { s } \\sum _ { i = 1 } ^ { a } \\pi _ { i } S _ { i } ^
{ \\frac { 1 } { 1 + s } } \\left [ \\log S _ { i } ^ { \\frac { 1 }
{ 1 + s } } - \\log A ( s ) \\right ] .</formula>
```

Experimental Setup

LoRA Parameters

- `r`: 16 (rank dimension)
- `lora_alpha`: 16 (scaling factor)
- `lora_dropout`: 0 (no dropout)
- `bias`: "none"

Fine-tuning Targets

- Vision layers: × Frozen
- Language layers: ✓ Trainable
- Attention modules: ✓ LoRA
- MLP modules: ✓ LoRA

Training Details

- Checkpoints saved every 100 steps
- Keep last 2 checkpoints
- Logging every 10 steps
- Max sequence length: 4,096
- FP16/BF16 precision: Auto-detect
- Random seed: 42

Training Configuration

- Batch size: 128
- Gradient accumulation: 1
- Max steps: 500
- Learning rate: 1e-4
- Scheduler: Cosine annealing
- Warmup steps: 20
- Optimizer: AdamW 8-bit
- Weight decay: 0.01

Fine-tuning Loss Curves

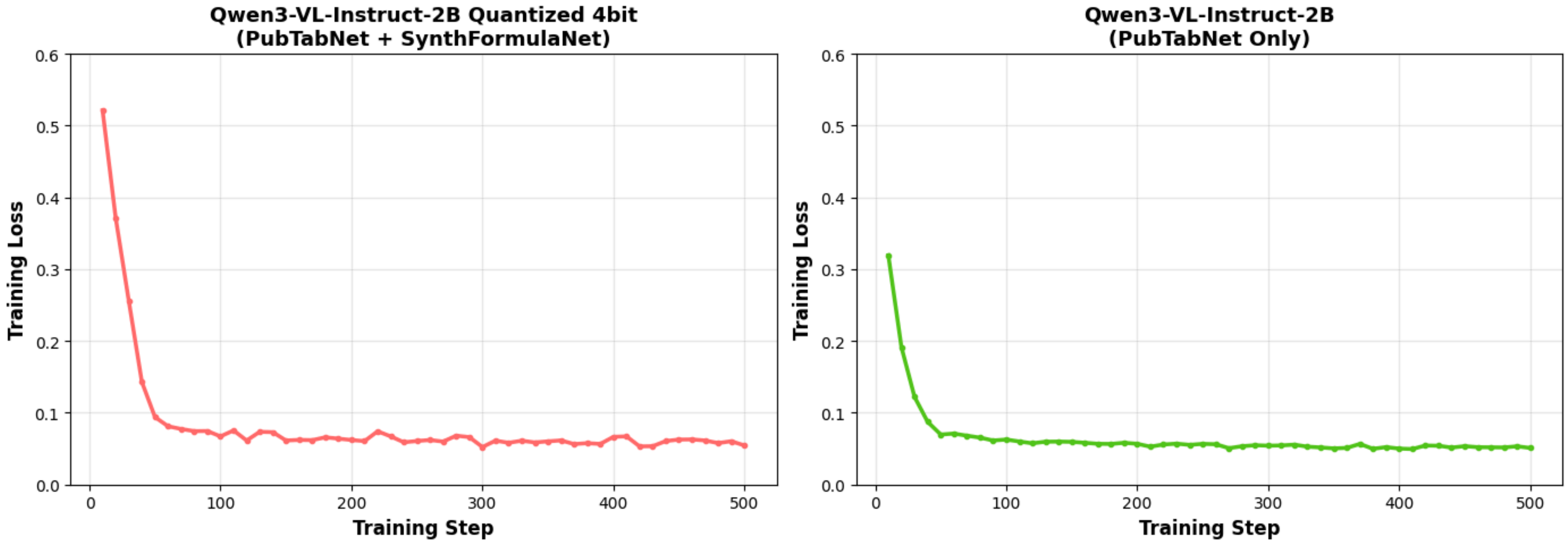


Figure 13: Finetuning using Qwen3-VL-Instruct-2B

Quantized + Combined Datasets

- Slower initial convergence
- Higher starting loss
- Noisier training curve
- Struggles with dual tasks

PubTabNet Only - No Quantization

- Faster convergence
- Lower starting loss
- Smoother training dynamics
- Better suited for task

Results: After finetuning

	text_block_Edit_dist_score	display_formula_CDM	table_TEDS	table_TEDS_structure_only	overall
MonkeyOCR-3B (baseline)	92.559000	84.070000	70.763000	74.704000	82.464000
InternVL3.5-1B-Flash	75.097000	38.145000	71.851000	81.624000	61.697667
Qwen3-VL-2B-Instruct	90.942000	80.677000	73.437000	79.425000	81.685333
Qwen3-VL-2B-Instruct-bnb-4bit	91.172000	82.096000	71.506000	78.633000	81.591333
Qwen3-VL-2B-Instruct-bnb-4bit (QLoRA pubtabnet-5k)	91.048000	81.900000	64.515000	84.783000	79.154333
Qwen3-VL-2B-Instruct-bnb-4bit (QLoRA pubtabnet-15k)	91.202000	83.708000	65.013000	85.833000	79.974333
Qwen3-VL-2B-Instruct-bnb-4bit (QLoRA pubtabnet+synthformula)	90.141000	19.069000	65.375000	86.973000	58.195000
Qwen3-VL-2B-Instruct (LoRA pubtabnet-15k)	91.791000	83.829000	65.454000	86.984000	80.358000

Fine-Tuning Error Analysis

Based on OmniDocBench benchmark evaluation, we identified three critical failure modes:

Chinese Text Handling

Fine-tuned model struggles with:

- Character recognition
- Encoding issues
- Mixed language documents

Root Cause: No Chinese samples in fine-tuning data

Formatting Inconsistency

Model fails to maintain:

- Layout structure
- Reading order
- Spatial relationships
- Hierarchical organization

Root Cause: Insufficient structural supervision during fine-tuning

Rotated Image Handling

Poor performance on:

- Rotated documents
- Skewed pages
- Non-standard orientations
- Extreme angles

Root Cause: Training data lacked orientation diversity

Error Analysis (after SFT) – Chinese tables confusion

Image

表 35 1935 年度宁夏省小学概况统计简表[3]

		省立	宁夏	宁朔	平罗	中卫	中宁	金积	灵武	盐池	豫旺	磴口	总计
学校数	完小	9	3	5	8	5	5	3	4	2	4	1	49
	初小	4	32	21	25	22	28	16	13	6	9	3	179
	合计	13	35	26	33	27	33	19	17	8	13	4	228
学生数	男	1846	2029	1410	2217	1826	2041	797	1038	287	547	131	14169
	女	550	61	73	201	202	347	111	51	82	122		1691
	合计	2396	2090	1483	2418	2028	2388	908	1089	369	560	131	15860
教职员	男	88	52	43	71	58	65	27	32	13	21	6	477
	女	16			1		1	4		1		1	24
	合计	104	52	43	72	58	66	31	32	14	21	7	501

1935 年，统计宁夏省总计学生 15860 人，其中仅有女生 1690 人，而男生 14169 人，差不多是女生的 10 倍。盐池县不仅女生只有

Ground truth

```
<html><body><table border="1" ><tr><td
colspan="2"></td><td>省立</td><td>宁夏</td><td>宁朔
</td><td>平罗</td><td>中卫</td><td>中宁</td><td>金积
</td><td>灵武</td><td>盐池</td><td>豫旺</td><td>磴口
</td><td>总计</td></tr><tr><td rowspan="3">学校数</
td><td>完小</td><td>9</td><td>3</td><td>5</
td><td>8</td><td>5</td><td>5</td><td>3</td><td>4</
td><td>2</td><td>4</td><td>1</td><td>49</td></
tr><tr><td>初小</td><td>4</td><td>32</td><td>21</
td><td>25</td><td>22</td><td>28</td><td>16</
td><td>13</td><td>6</td><td>9</td><td>3</
td><td>179</td></tr><tr><td>合计</td><td>13</
td><td>35</td><td>26</td><td>33</td><td>27</
td><td>33</td><td>19</td><td>17</td><td>8</
td><td>13</td><td>4</td><td>228</td></tr><tr><td
rowspan="3">学生数</td><td>男</td><td>1846</
td><td>2029</td><td>1410</td><td>2217</
td><td>1826</td><td>2041</td><td>797</
td><td>1038</td><td>287</td><td>547</td><td>131</
td><td>14169</td></tr><tr><td>女</td><td>550</
td><td>61</td><td>73</td><td>201</td><td>202</
td><td>347</td><td>111</td><td>51</td><td>8
```

cut-off.

Predicted

```
<html><body><table border="1" ><tr><td
rowspan="2"></td><td colspan="8"><b> 省立 </b></
td><td colspan="3"><b> 宁夏 </b></td><td
colspan="3"><b> 宁朔 </b></td><td colspan="3"><b>
平罗 </b></td><td colspan="3"><b> 中卫 </b></td><td
colspan="3"><b> 中宁 </b></td><td colspan="3"><b>
金积 </b></td><td colspan="3"><b> 灵武 </b></td><td
colspan="3"><b> 盐池 </b></td><td colspan="3"><b>
豫旺 </b></td><td colspan="3"><b> 磁口 </b></td><td
colspan="3"><b> 总计 </b></td></tr><tr><td><b> 学校
数 </b></td><td><b> 完小 </b></td><td><b> 9 </b></
td><td><b> 3 </b></td><td><b> 5 </b></td><td><b> 8
</b></td><td><b> 5 </b></td><td><b> 5 </b></
td><td><b> 3 </b></td><td><b> 4 </b></td><td><b> 2
</b></td><td><b> 4 </b></td><td><b> 1 </b></
td><td><b> 49 </b></td><td></td><td></td><td></
td><td></td><td></td><td></td><td></td><td></
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td><td></td><td></td><td></td><td></td><td></
td><td></td><td></td><td></t
```

cut-off

Error Analysis (after SFT) – Formula format

Image

$$
\begin{array}{rcl}
\displaystyle \binom{\alpha}{n} - \binom{\alpha-1}{n} & = & \displaystyle \frac{1}{n!}\alpha(\alpha-1)(\alpha-2)\dots(\alpha-n+1) \\
& & \displaystyle -\frac{1}{n!}(\alpha-1)(\alpha-2)\dots(\alpha-n) \\
& = & \displaystyle \frac{1}{n!}(\alpha-1)(\alpha-2)\dots(\alpha-n+1)(\alpha-(\alpha-n)) \\
& = & \displaystyle \frac{1}{(n-1)!}(\alpha-1)(\alpha-2)\dots(\alpha-n+1) \\
& = & \displaystyle \binom{\alpha-1}{n-1}.
\end{array}
$$

Ground truth

```
    \begin{array}{rcl}
\displaystyle \binom{\alpha}{n} -
\binom{\alpha-1}{n} & = & \displaystyle
\frac{1}{n!} \alpha(\alpha-1)
(\alpha-2)\dots(\alpha-n+1) \\
& & \displaystyle - \frac{1}{n!}
(\alpha-1)(\alpha-2)\dots(\alpha-n) \\
& = & \displaystyle \frac{1}{n!}
(\alpha-1)(\alpha-2)\dots(\alpha-n+1)
(\alpha - (\alpha-n)) \\
& = & \displaystyle \frac{1}{(n-1)!}
(\alpha-1)(\alpha-2)\dots(\alpha-n+1) \\
& = & \displaystyle \binom{\alpha-1}
{n-1}.
\end{array}
$$
```

Predicted

```
{'assistant':
'<loc_0><loc_0><loc_500><loc_500>( \
\alpha ) _ { n } - ( \\alpha - 1 ) _
{ n } = \\frac { 1 } { n ! } \\alpha ( \
\alpha - 1 ) ( \\alpha - 2 ) \\dots ( \
\alpha - n + 1 ) \\\\ - \\frac { 1 }
{ n ! } ( \\alpha - 1 ) ( \\alpha - 2 )
\\dots ( \\alpha - n ) \\\\ = \\frac
{ 1 } { n ! } ( \\alpha - 1 ) ( \\alpha
- 2 ) \\dots ( \\alpha - n + 1 ) ( \
\alpha - ( \\alpha - n ) ) \\\\ = \\frac
{ 1 } { ( n - 1 ) ! } ( \\alpha - 1 )
( \\alpha - 2 ) \\dots ( \\alpha - n +
1 ) \\\\ = ( \\alpha - 1 ) _ { n -
1 } .</formula>', 'source':
'SynthFormulaNet', 'user': '<formula>'}
```

)

Wins: Size, Speed

- **Size**
  - We reduced parameter usage by **1.5x**, by going from **3B** to **2B** parameters
  - We provide 4-bit quantized versions of the models while retaining **96%** overall performance.
- **Speed**
  - By developing a new infrastructure for inference, we achieved up to **2.4x** speedup in batch OCR, reducing OmniDocBench inference time from **1h56m** to only **48 minutes**.
  - This was done by developing a queueing mechanism, which batches multiple VLM calls and maximizes GPU utilization.

Conclusion

Key Findings & Outcomes

Fine-tuning Results

- Attempted LoRA and QLoRA fine-tuning on Qwen3-VL-Instruct-2B
- Achieved some improvements in specific metrics
- **Overall result:** Performance degraded in general evaluation due to limited data diversity, insufficient structural supervision

Best Overall Performing Approach

- **Model upgrade:** Qwen3-VL-Instruct-2B (no fine-tuning)
- **Speedup achieved:** 2.4× faster inference
- **Trade-off:** Maintained 99.0% of overall score while significantly reducing latency
- **If VRAM small:** Can use the 4-bit quantized version while retaining most of the performance and inference speed.

Lessons Learned

- Architectural improvements (model selection) can be better than fine-tuning on limited data
- Data quality and diversity critical for successful fine-tuning
- Ensure alignment of training data with benchmark formats

AI Interaction

Ideation & Strategy

- Brainstorming optimization approaches
- Evaluating trade-offs (speed vs. accuracy)
- Selecting backbone models

Dataset Discovery

- Locating suitable fine-tuning datasets
- Understanding PubTabNet structure
- Recommending domain-specific data

Experimental Configuration

- Proposing hyperparameters
- Setting up LoRA parameters

Implementation Support

- Code snippet generation
- Usage clarifications
- Debugging assistance
- Architecture modifications
- Better parallel implementation
- Plotting and error analysis tools

Future Work

- **Multilingual Fine-tuning Data**
  - Collect diverse Chinese and mixed-language samples
  - Improve cross-lingual generalization
- **Augmentation Strategies**
  - Synthetic rotations, skewing, and layout variations
  - Enhance robustness to document orientations
- **Hybrid Approaches**
  - Combine specialized models for different content types
  - Task-specific optimizations (text vs. tables vs. formulas)
- **Advanced Batching**
  - Dynamic batch size optimization based on block complexity
  - Adaptive resource allocation
- **Model Distillation**
  - Transfer knowledge from larger models
  - Maintain quality with further size reduction